

Notes: Diamond and Rajan (QJE, 2011)

Fear of Fire Sales, Illiquidity Seeking, and Credit Freezes

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	This is a theory paper: what the key objects are	4
2.2	Agents	4
2.3	Timeline	4
2.4	Assets and liabilities	5
2.5	Key frictions	5
2.6	What the model is designed to capture	5
3	Models and empirical specifications (all equations + interpretation)	6
3.1	Baseline setup	6
3.2	Buyers' bid price at date 0	6
3.3	Solvent bank's reservation price	6
3.4	Limited liability and the no-trade result	7
3.5	Extended model with securities, loans, and outside liquidity	7
3.6	Bank solvency condition in the extended model	8
3.7	Comparative statics: prices and lending	8
3.8	Runs and inefficiency	9
3.9	Numerical example	9
3.10	Extension 1: heterogeneity in banks' exposure to the shock	9
3.11	Extension 2: illiquidity risk versus pure fundamental risk	10
3.12	Forced date-0 asset sales	10
3.13	Other interventions	11
4	Identification and instruments (what makes the estimates credible)	11

4.1	This is a theory paper: what pins down the result	11
4.2	Why the no-trade result is not just adverse selection	11
4.3	Why limited liability is central	12
4.4	Why short-term debt matters	12
4.5	Why fear of fire sales affects ordinary lending	12
4.6	What assumptions do the heavy lifting	12
4.7	Relation to the literature	13
5	Figures	13
5.1	Figure I: prices of securities and liquidity	13
5.2	Figure II: date-0 lending	13
5.3	Read the figures together	14
6	Short summary	14
7	Conclusion	14
7.1	Core conclusion	14
7.2	Economic intuition	14
7.3	Takeaway	15

1 Big picture

- **Question.** Why do highly leveraged banks often refuse to sell illiquid assets even when early sales could protect them from future distress? How can the *fear of future fire sales* freeze the market for those assets today and simultaneously reduce credit to the rest of the economy?
- **Setting.** The paper develops a **theoretical model** of banks with **short-term liabilities**, **illiquid assets**, and **limited liability**. In the extended model, liquid buyers also choose between buying distressed securities and making ordinary term loans.
- **Core challenge.** In a crisis, one might think that weak banks should want to sell illiquid assets early, before prices collapse further. The paper shows that this intuition can fail because equity holders care about payoffs *conditional on survival*, and early sales partly transfer value to debt holders.
- **Main theoretical contribution.** The paper links three objects in one framework:
 - **future fire-sale risk**,
 - **banks' private incentive to retain or even accumulate illiquid assets**,
 - **and a current contraction in lending by healthy liquid buyers**.

The result is a model of **market freezes** driven by anticipated illiquidity rather than by pure fundamental risk alone.

- **Main mechanism.** If a bank would become insolvent when a future liquidity shock hits, its owners prefer not to sell the illiquid asset today. Selling early would insure the bank against the bad state, but much of that benefit goes to depositors or other debt holders. The bank therefore prefers to keep exposure to the upside in the survival state. In the paper's language, banks become **illiquidity seekers**.
- **Why it matters.** The model helps explain crisis behavior that otherwise looks puzzling:
 - weak institutions stop selling troubled assets,
 - liquid buyers hold cash back in anticipation of buying cheaply later,
 - and lending to the broader economy falls.

So the same force that freezes the asset market can also freeze credit.

- **Main results.**
 - If a bank remains solvent after a future liquidity shock, its date-0 reservation price equals buyers' bid price, so trade can occur.
 - If the bank would become insolvent after the shock, its date-0 asking price exceeds the market bid, so **no trade** occurs.
 - More potential liquidity demand, more short-term debt, or less outside liquidity lowers date-0 prices and lending.
 - Expected future fire sales raise the return to holding cash, which depresses current lending even by healthy liquid buyers.
 - Forced early asset sales or credible support that removes future fire sales can raise both asset prices and lending, but the design of intervention matters.

- **Policy message.** The problem is not only bad fundamentals. It is also the **institutional overhang** created by weak leveraged holders of illiquid assets. To restore credit, policy may need to move those assets into safer hands or otherwise eliminate the anticipated fire sale.
- **Economic takeaway.** A banking crisis can be self-reinforcing because the institutions most exposed to illiquidity have private incentives to hold on to illiquid assets, while potential buyers wait for even better opportunities later. This widens the fire sale and deepens the credit freeze.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no empirical dataset**. The paper is organized around the primitives of a banking and asset-market model.
- The key endogenous objects are:
 - the date-0 asset price P_0 ,
 - the date-1 fire-sale price P_1 ,
 - the quantity of ordinary lending at date 0,
 - and whether banks remain solvent or are run after a liquidity shock.

2.2 Agents

- **Illiquid banks.**
 - hold assets that mature at date 2,
 - finance themselves with short-term demandable debt or deposits of face value D ,
 - have limited-liability equity holders.
- **Depositors / short-term creditors.**
 - can withdraw at date 1 or wait until date 2,
 - require cash when they withdraw.
- **Liquid buyers.**
 - are not exposed to the liquidity shock,
 - start with cash θ ,
 - can buy distressed securities or make ordinary term loans.

2.3 Timeline

- **Date 0.**
 - banks may sell illiquid securities,

- liquid buyers decide how much cash to keep, how many securities to buy, and how much to lend.
- **Date 1.**
 - with probability q , a liquidity shock hits,
 - a fraction f of deposits withdraws,
 - banks sell securities and may liquidate loans to raise cash.
- **Date 2.**
 - assets and loans mature,
 - depositors are repaid,
 - residual value goes to bank equity and buyers.

2.4 Assets and liabilities

- In the **baseline model**, each bank holds one illiquid asset worth Z at date 2.
- In the **extended model**, a fraction β of bank assets is made up of securities that can be sold, and a fraction $(1 - \beta)$ is made up of bank loans that can be **recalled** / **liquidated** but not sold.
- Loans have face value Z at date 2. Their liquidation value at date 1 is heterogeneous and uniformly distributed between 0 and Z .
- Liquid buyers can alternatively make term loans at date 0. If the gross return on those loans is R , the volume of such lending is $I(R)$, with

$$I(1) = \bar{I}, \quad I'(R) < 0, \quad I''(R) > 0 \text{ for } R > 1.$$

2.5 Key frictions

- **Illiquidity.** The set of skilled buyers is limited and has finite cash. So when many banks need liquidity at once, asset prices collapse.
- **Short-term funding.** Deposits can be withdrawn before the asset matures, so banks may need to sell assets at bad prices.
- **Limited liability.** Bank equity holders do not fully internalize the value that early sales create for debt holders in bad states.
- **Funding does not fully reprice.** A central assumption is that the cost of deposits or short-term debt does not fully adjust ex ante to the bank's default risk. This is one reason the bank privately likes illiquidity more than society does.

2.6 What the model is designed to capture

- **Asset-market freeze:** no trade in troubled securities at date 0.
- **Credit freeze:** less ordinary lending at date 0 because buyers hoard cash for future fire sales.

- **Runs and inefficiency:** if banks become insolvent after the shock, they dump all securities and recall too many loans.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline setup

Each bank owns an asset that pays

$$Z > D$$

at date 2 and is financed by short-term deposits of face value D .

At date 1, with probability q , a liquidity shock arrives and a fraction f of deposits withdraws. If the bank waits until date 1 to sell assets, it must sell a fraction η_1 satisfying

$$\eta_1 Z P_1 = f D \quad \implies \quad \eta_1 = \frac{f D}{Z P_1}.$$

Interpretation.

- P_1 is the **fire-sale price** per unit of date-2 face value.
- The worse is P_1 , the larger is the fraction of the asset the bank must sell to meet withdrawals.

3.2 Buyers' bid price at date 0

Liquid buyers compare buying the asset at date 0 with holding cash and buying at date 1 if the shock hits. Their indifference condition implies

$$P_0^{\text{bid}} = \frac{1}{q \frac{1}{P_1} + (1 - q)}. \quad (1)$$

Interpretation.

- If the future fire-sale price P_1 is low, holding cash becomes attractive because buyers expect very high returns if they purchase later.
- Therefore anticipated future illiquidity depresses the **current** price P_0 .

3.3 Solvent bank's reservation price

If the bank is certain to remain solvent even when the shock hits, selling at date 0 simply replaces a possibly cheap future sale with a certain earlier sale. The bank is indifferent between selling at date 0 and waiting until date 1 when

$$P_0^{\text{ask}} = \frac{1}{q \frac{1}{P_1} + (1 - q)}. \quad (2)$$

Interpretation.

- When the bank is solvent in all states, its asking price equals the buyer's bid price.
- So there is no endogenous market freeze in this case.

3.4 Limited liability and the no-trade result

Suppose instead that the bank becomes insolvent if it does not sell assets before the shock arrives. Then equity holders value the asset mainly in the *survival* state. They prefer to hold the asset and keep the upside rather than sell early and strengthen depositors' position in the bad state.

The paper shows that there is no date-0 trade whenever the bank is insolvent after the shock and

$$fD > P_1[Z - (1 - f)D].$$

This yields:

Proposition 1. If the bank is insolvent at date 1 conditional on the liquidity shock and not having sold assets, it will never sell the asset at the bid price at date 0, even if by doing so it could remain solvent. No trade takes place at date 0.

Interpretation.

- This is the heart of the paper.
- The bank is effectively refusing to buy liquidity insurance because the insurance mostly protects debt holders.
- The result looks like a debt-overhang / risk-shifting logic, but the risk comes specifically from **future illiquidity**.

3.5 Extended model with securities, loans, and outside liquidity

Now let fraction β of bank assets be tradable securities and fraction $(1 - \beta)$ be loans that can be liquidated but not sold.

Liquid buyers hold cash θ and can make date-0 loans. If the implied gross return on date-0 lending must match the return on buying the illiquid asset, then lending equals

$$I\left(\frac{1}{P_0}\right).$$

Conditional on the liquidity shock, the bank liquidates all loans with liquidation value above $P_1 Z$. The total cash raised from loan liquidation is

$$(1 - \beta) \int_{P_1 Z}^Z x dx = (1 - \beta) \frac{Z}{2} (1 - P_1^2).$$

Cash supplied by loan liquidation plus cash held back by buyers must equal cash demanded by withdrawing depositors:

$$(1 - \beta) \frac{Z}{2} (1 - P_1^2) + \left[\theta - I\left(\frac{1}{P_0}\right) \right] = fD. \quad (3)$$

The date-0 and date-1 prices are linked by

$$P_0 = \frac{1}{q\frac{1}{P_1} + (1-q)}.$$

Interpretation.

- Equation (3) determines the fire-sale price from the economy's liquidity shortage.
- If buyers reserve more cash for future security purchases, less is lent at date 0.
- So the asset-market freeze and the credit freeze are jointly determined.

3.6 Bank solvency condition in the extended model

The bank remains solvent after the shock if the value of its remaining securities and loans is sufficient to pay the depositors who do not withdraw. The solvency condition is

$$P_1[\beta Z + (1-\beta)P_1 Z - (1-f)D] + \left\{ (1-\beta)\frac{Z}{2}(1-P_1^2) - fD \right\} \geq 0. \quad (4)$$

Interpretation.

- The first bracket is the discounted value of long-term assets net of remaining deposits.
- The curly bracket is the liquidity shortfall after loan liquidation.
- A lower P_1 makes both liquidity and solvency worse, which is why fire sales can trigger runs.

3.7 Comparative statics: prices and lending

Lemma 1.

- Higher liquidity demand f or higher bank debt D , and lower outside liquidity θ , reduce both P_0 and P_1 .
- A higher shock probability q lowers P_0 but raises P_1 .
- There are threshold values beyond which date-0 trading stops completely.

Proposition 2. An increase in f , D , or q , or a decrease in θ , reduces date-0 lending.

Interpretation.

- Higher expected future fire-sale returns make cash more valuable today.
- That raises the hurdle rate for ordinary lending.
- The especially interesting comparative static is q : a more likely future shock lowers P_0 because buyers want to preserve cash, even though it can raise P_1 because more precautionary cash is held back for the shock state.

3.8 Runs and inefficiency

If the bank becomes insolvent, it is run. Then it sells all securities for available cash and recalls all loans regardless of liquidation value.

Interpretation.

- Fire sales while the bank remains solvent are just transfers of liquidity.
- But insolvency causes an inefficient liquidation of real projects and a deeper collapse in lending.

3.9 Numerical example

The paper's baseline example sets

$$Z = 2, \quad \theta = 0.3, \quad f = 0.58, \quad q = 0.2, \quad \beta = 0.5,$$

and

$$I\left(\frac{1}{P_0}\right) = 0.3 \times 1.2^{1-\frac{1}{P_0}}.$$

The resulting equilibrium values are

$$P_0 = 0.354, \quad P_1 = 0.0988,$$

and date-0 lending equals

$$0.215.$$

Interpretation.

- The bank could sell about **24%** of its securities at date 0 and remain solvent.
- If it does not sell early, it must sell about **86%** of its securities after the shock and becomes insolvent.
- Even so, the bank prefers not to sell early because the expected value of equity is higher when it keeps the illiquid exposure.

3.10 Extension 1: heterogeneity in banks' exposure to the shock

Suppose that conditional on the aggregate shock, bank i is hit only with probability $\delta_i \leq 1$.

Proposition 3. If banks differ in δ_i , the ask price does not vary with δ_i when banks are solvent conditional on the shock hitting them, but it increases in δ_i when they are insolvent conditional on the shock.

Interpretation.

- If the bank remains solvent, lower exposure just means it can reinvest spare cash in the shock state, so the asking price is unaffected.
- If the bank would fail when hit, higher exposure makes the limited-liability put option more valuable, so the bank becomes even less willing to sell.

3.11 Extension 2: illiquidity risk versus pure fundamental risk

The paper also compares its mechanism with a world where date-1 prices are low because fundamentals are bad rather than because assets are illiquid.

Interpretation.

- If low future prices come from bad cash flows alone, buyers do *not* expect unusually high future returns from preserving cash.
- Then current lending need not collapse.
- This is one of the paper's key conceptual points: **fear of fire sales** is different from ordinary bad fundamentals.

3.12 Forced date-0 asset sales

The main policy proposition studies whether forced early sales can save banks.

Proposition 4. Let the bank be insolvent if it retains all securities until date 1 so that the solvency condition (4) is violated.

If there exists an $\eta_0 \in (0, 1)$ such that

$$(1 - \beta)P_1 Z P_1 + (1 - \beta)\frac{Z}{2}[1 - P_1^2] + \beta Z[\eta_0 P_0 + (1 - \eta_0)P_1] \geq (1 - f)D P_1 + fD \quad (5)$$

and

$$\beta Z \eta_0 P_0 \leq \left[\theta - I\left(\frac{1}{P_0}\right) \right], \quad (6)$$

then all banks can be made failsafe by forcing them to sell η_0 of their securities at date 0.

If that is impossible, only a fraction γ of banks may be saved, where γ , P_1 , and η_0 solve

$$\gamma \beta Z \eta_0 P_0 = \left[\theta - I\left(\frac{1}{P_0}\right) \right], \quad (7)$$

$$(1 - \gamma)\beta Z P_1 + \gamma \left\{ fD - (1 - \beta)\frac{Z}{2}[1 - P_1^2] \right\} = \left[\theta - I\left(\frac{1}{P_0}\right) \right], \quad (8)$$

and

$$(1 - \beta)P_1 Z P_1 + (1 - \beta)\frac{Z}{2}[1 - P_1^2] + \beta Z[\eta_0 P_0 + (1 - \eta_0)P_1] = (1 - f)D P_1 + fD. \quad (9)$$

Interpretation.

- Early forced sales make banks purchase liquidity insurance.
- They can raise both P_0 and P_1 by reducing how many banks dump securities after the shock.
- But if buyers' cash is too limited, forcing too many sales can reduce prices and backfire.

3.13 Other interventions

- **Voluntary sales / reverse auctions.** Authorities can buy assets directly, but then must pay for the banks' put option and may end up buying mostly from safer banks first.
- **Lending to liquid buyers.** Expanding buyers' financing effectively raises θ and can support prices, but may require subsidies or rents.
- **Lending to banks at date 1.** A credible commitment to provide liquidity when the shock hits can eliminate the fire sale, but may be politically difficult and may create ex ante moral hazard.
- **Closing some banks.** If all banks cannot be saved, shutting some down and removing their assets from the market can help the rest survive, but this requires asset valuation and resolution capacity.
- **Recapitalization.** Capital injections at the time of the shock can prevent runs and raise lending by keeping assets off the market, even if they do not induce early voluntary sales.
- **Ex ante regulation.** Capital or liquidity requirements can reduce illiquidity-seeking behavior before the crisis, but very tight regulation may also restrict financial intermediation.

4 Identification and instruments (what makes the estimates credible)

4.1 This is a theory paper: what pins down the result

- There is no empirical identification problem here. The relevant question is what **analytically** pins down the equilibrium.
- The core result comes from combining:
 - **finite outside liquidity,**
 - **short-term withdrawal risk,**
 - **limited liability,**
 - and a funding structure whose cost does not fully reprice with risk.
- These assumptions generate an endogenous wedge between what society would want weak banks to do and what bank equity privately wants to do.

4.2 Why the no-trade result is not just adverse selection

- The paper does **not** rely on classic Akerlof-style asymmetric information among the skilled buyers and sellers.
- Instead, the freeze comes from a forward-looking asset-pricing logic:
 - buyers hold back cash because future fire sales offer high returns,
 - weak banks hold back assets because selling early strengthens debt in bad states.
- This makes the mechanism conceptually different from lemons-based market freezes.

4.3 Why limited liability is central

- If banks fully internalized payoffs in all states, they would value the insurance provided by early sales.
- With limited liability, equity cares mostly about the states in which the bank survives.
- That is why banks can rationally choose to remain exposed to illiquidity risk even when the social consequences are bad.

4.4 Why short-term debt matters

- The asset is dangerous not only because it is illiquid, but because illiquidity interacts with **withdrawable funding**.
- A bank financed entirely with long-term liabilities would not face the same forced-sale problem even if it held the same asset.
- This is one of the paper's deepest insights: the same asset can be much riskier in one balance-sheet structure than in another.

4.5 Why fear of fire sales affects ordinary lending

- Liquid buyers choose between distressed-asset purchases and regular term lending.
- When expected returns from future fire sales rise, cash becomes more valuable and the required return on lending rises too.
- So the crisis spills over from the market for distressed securities to the rest of credit markets.

4.6 What assumptions do the heavy lifting

- risk neutrality,
- a three-date structure,
- a common liquidity shock,
- finite liquidity held by skilled buyers,
- deposits or short-term debt that do not fully reprice with default risk,
- limited liability of bank equity,
- a reduced-form outside lending schedule $I(R)$.

These assumptions keep the model simple, but they are also exactly what makes the mechanism transparent.

4.7 Relation to the literature

- Relative to **Jensen–Meckling** and **Myers**, the paper is a debt-overhang / risk-shifting story in which the key state variable is **illiquidity** rather than only default risk from fundamentals.
- Relative to **Shleifer–Vishny**, prices are shaped by limited buyer liquidity.
- Relative to **Diamond and Rajan (2005)**, the paper emphasizes how anticipated future fire sales distort today's asset trade and today's lending.
- Relative to pure lemons models, the freeze here is not driven by increasing private information over time.

5 Figures

5.1 Figure I: prices of securities and liquidity

- The figure plots date-0 and date-1 security prices against the amount of available outside liquidity θ .
- When liquidity is very scarce, banks fail conditional on the shock, both P_0 and P_1 are very low, and the market is effectively frozen.
- Once liquidity crosses a threshold, the banks remain solvent conditional on the shock, and both prices jump upward.
- The main lesson is **nonlinearity**: small liquidity improvements below the solvency threshold do little, but crossing the threshold changes the equilibrium sharply.

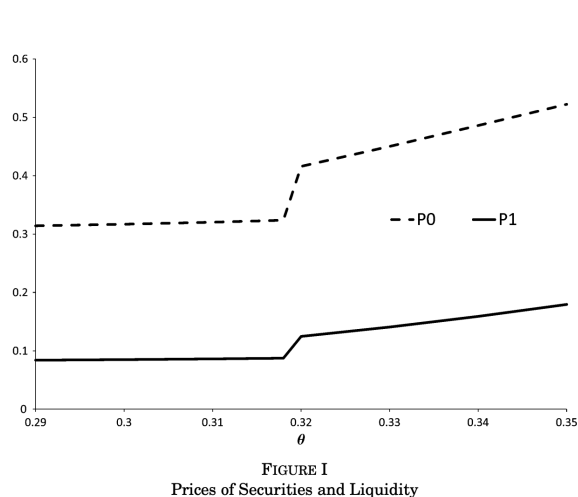


FIGURE I
Prices of Securities and Liquidity

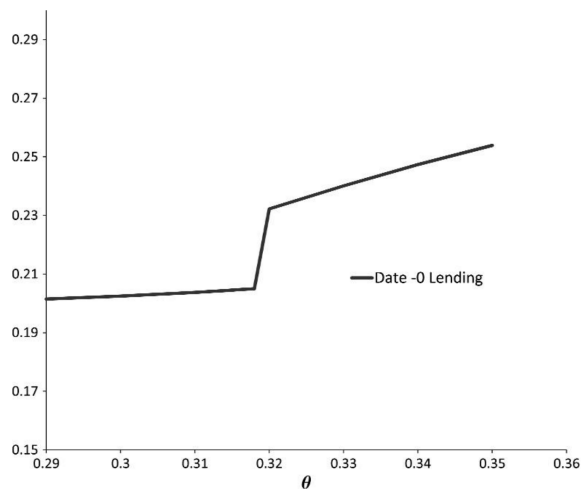


FIGURE II
Date-0 Lending

5.2 Figure II: date-0 lending

- The figure plots ordinary date-0 lending against outside liquidity θ .

- Lending stays depressed when the market still anticipates future fire sales.
- Once the banking system becomes solvent conditional on the shock, the expected return to hoarding cash falls and date-0 lending rises sharply.
- The key message is that **asset-market repair and credit recovery are tightly linked**.

5.3 Read the figures together

- Figure I shows the asset-pricing channel.
- Figure II shows the credit-supply channel.
- Together they illustrate the paper's central idea: the same expected future fire sale that lowers current asset prices also lowers current lending.

6 Short summary

- **Method.** A theoretical model of banks, illiquid assets, short-term funding, and liquid outside buyers.
- **Mechanism.** Weak banks refuse to sell illiquid assets because early sales insure debt holders in bad states, while buyers hold back cash to exploit future fire sales.
- **Main result.** Anticipated future fire sales can freeze current trading and reduce ordinary lending today.
- **Policy implication.** Restoring credit may require removing illiquid assets from weak leveraged institutions or otherwise eliminating the future fire-sale overhang.

7 Conclusion

7.1 Core conclusion

- The paper shows that privately optimal behavior by distressed banks can make a crisis worse.
- Banks with short-term liabilities and limited liability may *seek* illiquidity rather than shed it.
- Anticipated future fire sales can therefore produce a current market freeze and a current credit freeze.

7.2 Economic intuition

- Early asset sales look socially prudent because they reduce the chance of future insolvency.
- But from the viewpoint of bank equity, selling early is partly a transfer to debt holders.
- So weak banks prefer to gamble on survival states, while healthy buyers hold cash back in order to buy later at distressed prices.

7.3 Takeaway

The paper's main lesson is that **illiquidity is not just a property of assets**. It is a property of the interaction between **asset liquidity**, **balance-sheet structure**, and **the distribution of cash in the economy**. Once a crisis makes future fire sales plausible, private incentives can push the system toward less trade, less lending, and deeper fragility. That is why cleaning up the financial system can be necessary not only to save weak banks but also to restart credit.